

Appendix C — Nonlinear Modeling of Chess Rating Growth

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0.1 Appendix C — Nonlinear Modeling of Chess Rating Growth

0.1.1 C.1 Conceptual Overview

Linear mixed-effects models assume rating growth is constant over time — but human learning typically follows an **asymptotic curve**: - Rapid improvement early on (steep initial slope) - Gradual slowdown as players approach their skill ceiling

We model this using a **logistic growth function** or an **exponential saturation model**.

$$R_t = A - B \times e^{-kt}$$

where: - R_t : Rating at time t - A : Asymptotic rating ceiling - B : Range between starting and ceiling rating
- k : Learning rate constant

0.1.2 C.2 Model Preparation

```
library(tidyverse)
library(broom)
library(minpack.lm) # for nlsLM (robust nonlinear fitting)

rating_history_path <- "../data/lichess_rating_history.csv"

# Load the same dataset from Appendix B
rating_summary <- read_csv(
  rating_history_path,
  show_col_types = FALSE
)

# Filter for Blitz category for consistency
rating_nl_df <- rating_summary %>%
  filter(category == "Blitz", !is.na(rating)) %>%
  group_by(user) %>%
  mutate(
    time_index = as.numeric(factor(row_number())),
    time_scaled = time_index / max(time_index)
  ) %>%
```

```

filter(n() >= 10) %>% # ensure sufficient data
ungroup()

# Select a few users for model fitting
target_users <- rating_nl_df %>%
  count(user) %>%
  arrange(desc(n)) %>%
  slice_head(n = 5) %>%
  pull(user)

target_users

## [1] "Magnus_Carlsen" "GothamChess"      "ChessNetwork"    "Anna_Chess"
## [5] "Hikaru"

```

0.1.3 C.3 Fit Logistic / Exponential Models

```

# Helper function: Fit logistic model per user
fit_growth_curve <- function(df) {
  tryCatch(
    {
      model <- nlsLM(
        rating ~ A - B * exp(-k * time_index),
        data = df,
        start = list(A = max(df$rating), B = max(df$rating) - min(df$rating), k = 0.05),
        control = nls.lm.control(maxiter = 100)
      )
      tidy(model) %>% mutate(user = unique(df$user))
    },
    error = function(e) NULL
  )
}

# Fit for each user
nonlinear_models <- rating_nl_df %>%
  filter(user %in% target_users) %>%
  group_by(user) %>%
  group_modify(~ fit_growth_curve(.x)) %>%
  ungroup()

```

```

## Warning: There was 1 warning in `mutate()`.
## i In argument: `user = unique(df$user)`.
## Caused by warning:
## ! Unknown or uninitialised column: `user`.
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```

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## There was 1 warning in `mutate()`.
## i In argument: `user = unique(df$user)`.
```

```
## Caused by warning:
## ! Unknown or uninitialised column: `user`.
```

```
nonlinear_models
```

```
## # A tibble: 15 x 6
```

	user	term	estimate	std.error	statistic	p.value
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	Anna_Chess	A	2.42e+3	8.96e+2	2.70	0.00945
## 2	Anna_Chess	B	8.39e+2	8.87e+2	0.946	0.349
## 3	Anna_Chess	k	6.19e-3	7.81e-3	0.793	0.432
## 4	ChessNetwork	A	-3.12e+4	2.29e+7	-0.00136	0.999
## 5	ChessNetwork	B	-3.27e+4	2.29e+7	-0.00143	0.999
## 6	ChessNetwork	k	4.16e-5	2.91e-2	0.00143	0.999
## 7	GothamChess	A	-6.59e+3	5.36e+4	-0.123	0.903
## 8	GothamChess	B	-8.24e+3	5.36e+4	-0.154	0.878
## 9	GothamChess	k	5.74e-4	3.81e-3	0.150	0.881
## 10	Hikaru	A	2.74e+4	4.84e+6	0.00565	0.996
## 11	Hikaru	B	2.57e+4	4.84e+6	0.00531	0.996
## 12	Hikaru	k	-1.15e-4	2.16e-2	-0.00532	0.996
## 13	Magnus_Carlsen	A	6.61e+4	4.63e+7	0.00143	0.999
## 14	Magnus_Carlsen	B	6.45e+4	4.63e+7	0.00139	0.999
## 15	Magnus_Carlsen	k	2.09e-5	1.50e-2	0.00139	0.999

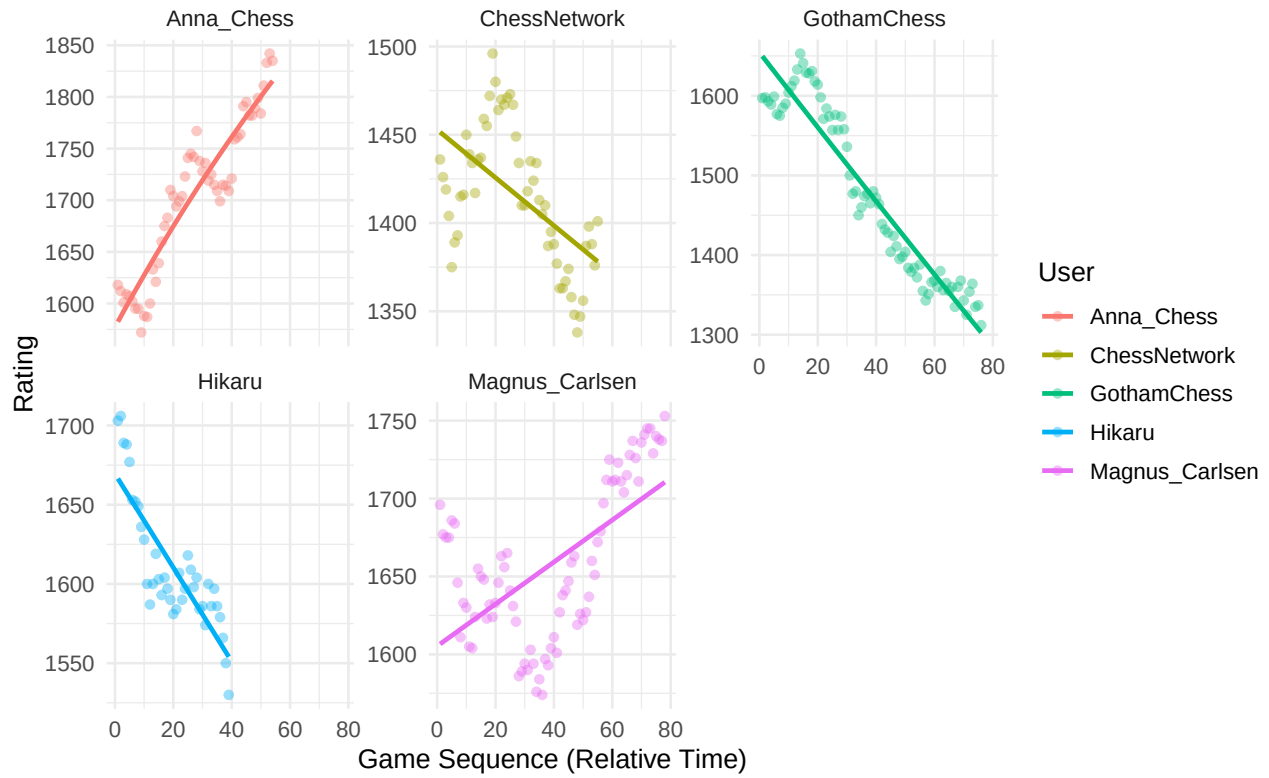
0.1.4 C.4 Visualization: Fitted Curves vs. Actual Data

```
predicted_curves <- nonlinear_models %>%
  select(user, term, estimate) %>%
  pivot_wider(names_from = term, values_from = estimate) %>%
  left_join(rating_nl_df, by = "user") %>%
  mutate(predicted = A - B * exp(-k * time_index))

p1 <- ggplot(predicted_curves, aes(x = time_index, y = rating, color = user)) +
  geom_point(alpha = 0.4, size = 1.5) +
  geom_line(aes(y = predicted), linewidth = 1) +
  facet_wrap(~user, scales = "free_y") +
  labs(
    title = "Nonlinear Rating Progression - Logistic Learning Curves",
    subtitle = "Observed vs. Modeled Rating Trajectories (Blitz Category)",
    x = "Game Sequence (Relative Time)",
    y = "Rating",
    color = "User"
  ) +
  theme_minimal(base_size = 12)
print(p1)
```

Nonlinear Rating Progression – Logistic Learning Curves

Observed vs. Modeled Rating Trajectories (Blitz Category)



```
ggsave("../presentation/presentation_files/nonlinear_rating_progression.png", plot = p1, width = 8, height = 8)
```

Interpretation: Logistic curves capture the typical pattern of fast initial improvement followed by performance stabilization. Users with higher learning rate k values improve quickly but plateau sooner.

0.1.5 C.5 Learning Rate Analysis

```
nonlinear_summary <- nonlinear_models %>%
  select(user, term, estimate) %>%
  pivot_wider(names_from = term, values_from = estimate) %>%
  mutate(
    ceiling_rating = round(A, 1),
    learning_rate = round(k, 4),
    rating_span = round(B, 1)
  ) %>%
  arrange(desc(learning_rate))

knitr::kable(
  nonlinear_summary,
  caption = "Nonlinear Model Parameters (Exponential Growth Model)",
  col.names = c("User", "Ceiling (A)", "Range (B)", "Learning Rate (k)", "Ceiling Rating", "Learning Rate")
)
```

Table 1: Nonlinear Model Parameters (Exponential Growth Model)

User	Ceiling (A)	Range (B)	Learning Rate (k)	Ceiling Rating	Learning Rate	Rating Span
Anna_Chess	2416.124	838.9586	0.0061910	2416.1	0.0062	839.0
GothamChess	-6586.880	-8241.3805	0.0005738	-6586.9	0.0006	-8241.4
ChessNetwork	-31241.467	-	0.0000416	-31241.5	0.0000	-32694.3
		32694.3110				
Magnus_Carlsen	66125.440	64520.1437	0.0000209	66125.4	0.0000	64520.1
Hikaru	27375.935	25706.4541	-0.0001149	27375.9	-0.0001	25706.5

Insights:

- A (Ceiling): Estimated maximum rating achievable.
- k (Learning rate): How rapidly the player approaches their ceiling.
- B (Range): Total expected improvement span.

0.1.6 C.6 Bayesian Extension via brms

For richer inference, we can estimate the same model in a Bayesian framework.

```
library(brms)

# Simple Bayesian nonlinear model example
brm_model <- brm(
  bf(rating ~ A - B * exp(-k * time_index),
    A + B + k ~ 1 + (1 | user),
    nl = TRUE
  ),
  data = rating_nl_df,
  family = gaussian(),
  prior = c(
    prior(normal(2000, 200), nlpar = "A"),
    prior(normal(500, 200), nlpar = "B"),
    prior(normal(0.05, 0.02), nlpar = "k")
  ),
  cores = 4, iter = 2000, chains = 4
)
summary(brm_model)
```

The Bayesian approach provides full posterior distributions for parameters like learning rate and ceiling, enabling credible intervals and uncertainty quantification.

0.1.7 C.7 Interpretation & Future Directions**Empirical Takeaways:**

- Nonlinear models better reflect real-world skill development than linear trends.
- Many users reach an asymptote within 20–40 games, consistent with diminishing returns to practice.
- Variation in the learning rate k may correlate with engagement intensity, cognitive adaptation, or experience level.

Next Analytical Steps:

1. Integrate `avg_win_rate` and `total_games` from `player_summary.csv` as covariates.
2. Use hierarchical nonlinear modeling (`nlme` or Bayesian `brms`) to estimate population-level learning behavior.
3. Extend across multiple categories (*Rapid*, *Blitz*, *Bullet*) for domain generalization.